Practical Examples of Quant Strategies

Alexander Alekseev

STYLES OF TRADING

- Portfolio Management
  - Hedge Funds, Long-only asset managers
  - Prohibited on the trading floor (Volker rule)
- Market Making
  - Provide bid/offer in the markets for other participants to trade against your quotes
  - Banks, Hedge Funds, prop. shops
- Arbitrage Trading
  - Identifying mispricing in the markets and trying to fill the gap
  - Usually spread-crossing trading.
  - Historically was done within the Banks, now moved to prop shops.
- Optimal Execution
  - Traditional banks' businesses
  - Offer clients (e.g. your pension fund) to buy/sell assets on their behalf.

BASE LAYER LOSE MORE MONEY DURING CRSIS

- Base Layer: Pension fund are the main source of capital 100T pension fund use simple strategy, ex: buy and hold some company's share for 5%, and they have impact on the board

- Second Layer: Long-only Asset Manager
ex: BlackRock

- Third Layer: Banks, Advisors
Work as an intermediate and help AM to manage the money. They should not do proprietary trading(trading for their own interest and take risk)

- Fourth Layer: Hedge Fund, Specialized fund - 5M
Like millenium and Balyasny

- fifth Layer: Prop shops
Like XTX, citidel security

- Model – based investment decisions
- An optimal allocation of investment across assets that takes into account:
  - Asset Valuation Model-- e.g. stock price prediction
  - Risk Model-- a model predicting expected correlations between the assets and the total risk of the portfolio
  - Objective Function-- what is that we want to maximize? What are the constraints?
  - A Portfolio optimization algorithm
- Usually relatively long holding periods
- Used by asset managers and hedge funds
- Banks apply the same models to manage inventories in a systematic manner

PORTFOLIO MANAGEMENT

- The core of Banks' trading businesses
- Moving more and more to electronic platforms
- Sales trading roles are disappearing
- Electronic trading is taking over
  - Cash equities: >95% of volume is electronic
  - FX: >80% if electronic
  - Futures/ETFs: >95 % is electronic
  - Options US: >95% electronic
  - Government Bonds: >95% is electronic
  - Companies Debt: moving to electronic platforms now
  - Structured products – OTC, all manual.
- Good High Frequency trading

MARKET MAKING

- Provide bid and offer spreads in the markets
- Capture spreads
- Deal with negative selection (your are filled only when the market goes against you)
- Predict Short term prices to skew quotes
- Predict Medium term prices to hold inventories around which you will be quoting
- Model risks to measure when and how aggressively you want to close our your inventories

MARKET MAKING

ARBITRAGE TRADING

- Examples of arbitrage situations
  - Two instruments that are trading in the market are linked by a formula-- e.g. a stock index and the future on the index (index arbitrage)
  - Two instruments are highly correlated and deviations from this correlation pattern can be explored (pair trading)
  - The same instrument is traded in two venues, e.g. on a local exchange and MTF. If bid on one is higher than the offer on the other you have a 'back to back' arbitrage opportunity.
- Difficult to explore - high cost of entry in these businesses, speed, high volumes, expensive hardware, tight models.
- Historically was originated in French banks, currently are mostly traded by highly specialized prop. shops.
- Regulators are targeting this specific area as the most prone to market manipulation.

LINK BETWEEN ARBITRAGE TRADING AND MARKET MAKING

- Let's look into an example of Index Arbitrage:
  - The idea is if the future price deviates from the it's theoretical value, it creates and arbitrage opportunity.
  - Future is overpriced ('rich') compared to the underlying index, you sell the future and buy the index
- How do you execute this trade?
  - Cross on future, cross on cash (regulator don't like that and it is inefficient)
  - Make market on future, cross on cash (classical Index Arb, it's fine with all regulations)
  - Make market on future, make markets on both and create market pressure to move the two instrument back inline together --- this what regulator should encourage.

LINK BETWEEN ARBITRAGE TRADING AND MARKET MAKING

- Arbitrage trading looks more and more like market making
- You can move two co-integrated instruments inline with each other by crossing the spreads or by making markets
- Market making engines can create market pressures to keep the instruments inline, not crossing the spreads
- This is more efficient (don't pay ½ spread)
- More difficult from the modelling prospective, but doable:)
- Pure arbitrage strategies become less and less profitable and market making engines drive them out of the business

- Traditional Banks' business
- Client comes with an order – e.g. buy 1mn shares of company X.
- The bank loads the order to an automated execution algorithm.
- This algorithm slices the order into small parts and slowly executes in the market
- It tries to reduce the 'market impact' of the order by
  - slicing the order
  - Utilizing short-term alphas to identify the optimal time to place these slices in the market
  - Internalizing the flow between different clients
  - Committing capital to hold the order on its own books until the opportune moment arrives

OPTIMAL EXECUTION

- Systematic approach to modelling and trading
- Key advantages of systematic trading:
  - Use models to take calculated risks
  - You can back test your strategy if it is systematic
  - You can do performance evaluation and identify weaker/stronger parts of your investment process
- The key problem with discretionary trading – you can't repeat the same action several times
- Law of large numbers –
  - if you come to a casino, make one big bet
  - If you invest, and you believe in your model, split your investment into small bets

COMMON PARTS OF ALL STYLES

- The key part of the investment process
- Diversity of Alphas:
  - Balance-sheet based information
  - Analysts forecasts
  - Fama-French type of f quant factors
  - Stock-price based information
  - Market sentiment predictions

VALUATION MODEL

sharpe of 1 would be good enough as self project

- LARS, Lasso
  - Select a statistical model that linearly combines several predictors
  - Penalty function which depends on the number of factors & the weights
  - Reduces the chances of overfitting due to multicollinearity
- Support Vector machines
  - Linear - find a hyperplane that is best to classify your observations
  - Non-linear – use kernels
- Neural Networks
  - Take a graph (math) with various connected nodes
  - Take a path from one node to another, one node is input (current stock price) of your model, the end one – output (e.g. how much your invest)
  - Train the network by eliminating implausible paths between the nodes
  - Once the network is trained, for every input (node) you would have a path leading to some node with optimal output (another node)

VALUATION MODEL, COMBINE ALPHAS

RISK MODELS

- Try to predicted covariance between instruments
- Commonly used- Barra, Axioma
  - Quant factor based models
  - Good coverage
  - Good stats behind calculations
- Intraday models are difficult due to spread noise distorting estimates. One of the successful attempts- JP Morgan & New Economic School
- Good academic papers on de-noising tick data (Roel Oomen, Deutsche Bank)
- PCA, simple estimates do not work

SOFTWARE

- Commonly used software:
  - Modelling
    - Matlab, R, python
  - Trading
    - C++
    - Java is almost not used anymore
  - Optimizers
    - Mosek
    - Small problems with simple constraints can be solved by Matlab optimizer

SKILLS THE INDUSTRY NEEDS

- Computers replace manual traders
- Most of the growth happens in the development and quant areas
- The line between trader/ quant/ developer is disappearing
- Ideal profile is a person who understand all bits of the process and specializes in one of the components, e.g.:
  - A trader who has strong quantitative background
  - A quant who can develop C++ code and verify dev. Implementation
  - A C++ developer who understands quant models
- Learn the finance theory and how to train computers do the job for you ☺

- Focus on efficiency
  - Automate all parts of the investment process
  - Reduce costs of infrastructure, headcount
  - Internalization of flow
  - Centralization of inventory management functions
- Market Making engines are taking over other businesses
  - Replacing manual functions
  - Driving the profit margins for arbitrage strategies down
  - Replacing Execution strategies by proving optimal way to enter/exit form the positions
- Focus on R&D
  - More different types of alphas feeding the engines
  - Better model selection techniques
  - Better coverage

TRENDS IN THE INDUSTRY